第七章 复数

Chapter 7 Complex Numbers

7.1 复数的概念 Concepts of Complex Numbers

核心知识归纳 Core Knowledge Summary

1. **数系扩充与虚数单位 Number System Extension and Imaginary Unit**

- 扩充背景:为解决方程 $x^2+1=0$ (或 $x^2+5=0$)在实数集中无解的问题,引入虚数单位 i 。

  Extension Background: To solve the problem that the equation $x^2+1=0$ (or $x^2+5=0$) has no solution in the set of real numbers, the imaginary unit i is introduced.

- 规定: $i^2=-1$; i 可与实数进行四则运算,运算律保持不变。

  Stipulation: $i^2=-1$; i can perform four arithmetic operations with real numbers, and the operational laws remain unchanged.

2. **复数的定义与分类 Definition and Classification of Complex Numbers**

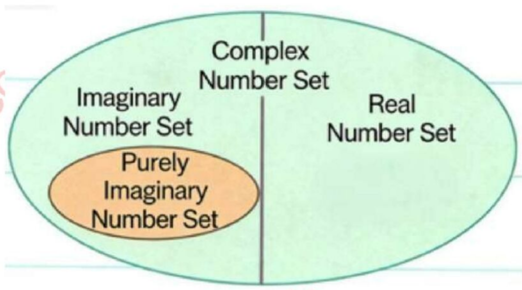
- 定义:形如 $z=a+bi$ (a 、 $b\in R$)的数叫做复数,其中:

  Definition: A number of the form $z=a+bi$ (a , $b\in R$) is called a complex number, where:

  - a 为实部($\mathrm{Re}(z)=a$), b 为虚部($\mathrm{Im}(z)=b$);

    a is the real part ($\mathrm{Re}(z)=a$), and b is the imaginary part ($\mathrm{Im}(z)=b$);

  - 全体复数构成复数集 $C=\{a+bi\mid a,b\in R\}$,实数集 R 是 C 的真子集 ($R\subsetneq C$)。



    All complex numbers form the set of

complex numbers $C = a + bi \mid a, b \in R$, and the set of real numbers R is a proper subset of C ($R \subsetneqq C$).

- 分类:Classification:
  - 实数: $b = 0$ (如 3 、 $-\sqrt{2}$ 、 0);

    Real Numbers: $b = 0$ (e.g., 3 , $-\sqrt{2}$, 0);
  - 虚数: $b \neq 0$ (如 $2 + 3i$ 、 $-0.2i$);

    Imaginary Numbers: $b \neq 0$ (e.g., $2 + 3i$, $-0.2i$);
  - 纯虚数: $a = 0$ 且 $b \neq 0$ (如 i 、 $\sqrt{3}i$)。

    Pure Imaginary Numbers: $a = 0$ and $b \neq 0$ (e.g., i , $\sqrt{3}i$).
- 关键:若两个复数有大小关系,则它们必为实数;虚数不能比较大小。

  Key Point: If two complex numbers have a size relationship, they must be real numbers; imaginary numbers cannot be compared in size.

3. 复数相等的充要条件 Necessary and Sufficient Condition for Equality of Complex Numbers

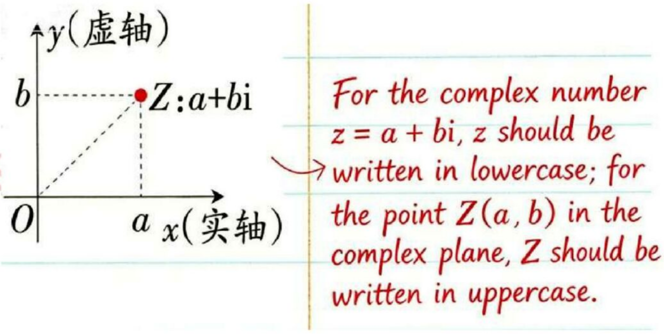
- 设 $z_1 = a + bi$, $z_2 = c + di$ (a 、 b 、 c 、 $d \in R$),则 $z_1 = z_2 \Leftrightarrow a = c$ 且 $b = d$ 。

  Let $z_1 = a + bi$ and $z_2 = c + di$ (a , b , c , $d \in R$), then $z_1 = z_2 \Leftrightarrow a = c$ and $b = d$.
- 前提: a 、 b 、 c 、 d 必须为实数,否则不能直接等同实部和虚部。

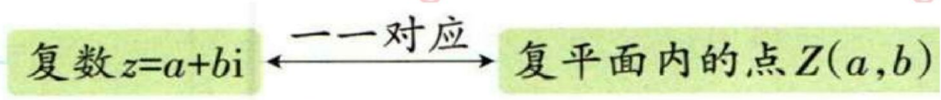
  Prerequisite: a , b , c , and d must be real numbers; otherwise, the real parts and imaginary parts cannot be directly equated.

4. 复数的几何意义 Geometric Meaning of Complex Numbers

- 复平面:建立直角坐标系, x 轴为实轴(表示实数), y 轴为虚轴(除原点外表示纯虚数);

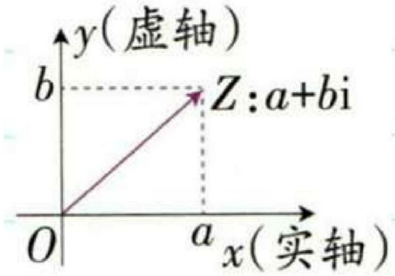


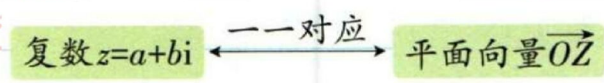
复数 $z=a+bi$ 与复平面内的点 $Z(a,b)$ 一一对应(点的坐标是 (a,b) ,而非 (a,bi))。



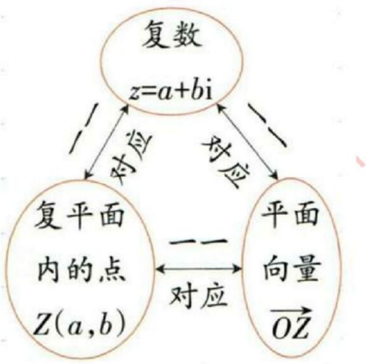
Complex Plane: Establish a rectangular coordinate system, the x -axis is the real axis (representing real numbers), and the y -axis is the imaginary axis (representing pure imaginary numbers except the origin); the complex number $z=a+bi$ is in one-to-one correspondence with the point $Z(a,b)$ in the complex plane (the coordinates of the point are (a,b) , not (a,bi)).

- 向量表示:复数 $z=a+bi$ 与复平面内以原点 O 为起点、 $Z(a,b)$ 为终点的向量 $\overrightarrow{OZ}$ 一一对应(实数 0 与零向量对应)。





Vector Representation: The complex number $z=a+bi$ is in one-to-one correspondence with the vector $\overrightarrow{OZ}$ in the complex plane with the origin O as the starting point and $Z(a,b)$ as the endpoint (the real number 0 corresponds to the zero vector).



5. 复数的模与共轭复数 Modulus and Conjugate Complex Number of Complex Numbers

- 模(绝对值):向量 $\overrightarrow{OZ}$ 的长度,记作 $|z|$ 或 $|a+bi|$,即 $|z|=\sqrt{a^2+b^2}$; 实数的模等于其绝对值。

  Modulus (Absolute Value): The length of the vector $\overrightarrow{OZ}$, denoted as $|z|$ or $|a+bi|$, i.e., $|z|=\sqrt{a^2+b^2}$; the modulus of a real number is equal to its absolute value.

- 共轭复数:

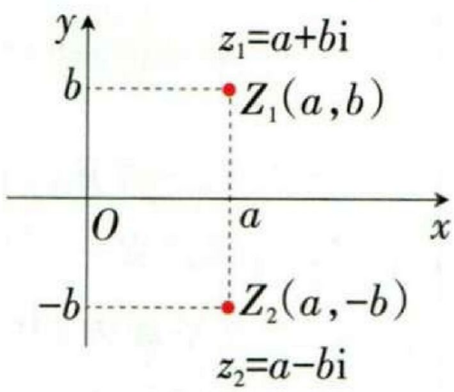
  Conjugate Complex Number:

  - 定义:实部相等、虚部互为相反数的复数,记作 $\bar{z}$,即 $\overline{a+bi}=a-bi$; 实数的共轭复数是其本身。

    Definition: A complex number with equal real parts and opposite imaginary parts, denoted as $\bar{z}$, i.e., $\overline{a+bi}=a-bi$; the conjugate complex number of a real number is itself.

  - 几何意义:复平面内对应的点关于实轴对称。

    Geometric Meaning: The corresponding points in the complex plane are symmetric about the real axis.



  - 性质: $\overline{(\bar{z})}=z$; $z+\bar{z}=2a$ (实数); $z-\bar{z}=2bi$ (纯虚数或 0); $z\cdot\bar{z}=|z|^2=a^2+b^2$ (实数)。

    Properties: $\overline{(\bar{z})}=z$; $z+\bar{z}=2a$ (real number); $z-\bar{z}=2bi$ (pure imaginary number or 0); $z\cdot\bar{z}=|z|^2=a^2+b^2$ (real number).

易错点解析 Common Mistakes Analysis

1. 虚部的定义误区:复数 $z=a+bi$ 的虚部是 b (实数),而非 bi (如 $2+3i$ 的虚部是 3 ,不是 $3i$)。

Misunderstanding the definition of the imaginary part: The imaginary part of the complex number $z=a+bi$ is b (a real number), not bi (e.g., the imaginary part of $2+3i$ is 3 , not $3i$).

2. 复平面的坐标误区:虚轴的单位长度是 1 ,而非 i ; 点 $Z(a,b)$ 的纵坐标是 b ,不是 bi 。

   Misunderstanding the coordinates of the complex plane: The unit length of the imaginary axis is 1 , not i ; the ordinate of the point $Z(a,b)$ is b , not bi .

3. 复数相等的条件误区:忽略 a 、 b 、 c 、 d 为实数的前提,直接等同虚部和实部。

   Misunderstanding the condition for equality of complex numbers: Ignoring the prerequisite that a , b , c , and d are real numbers, and directly equating the imaginary parts and real parts.

4. 共轭复数的性质误区: $z\cdot\bar{z}=|z|^2$ 是实数,但 $|z_1z_2|=|z_1||z_2|$, $|z_1+z_2|\le|z_1|+|z_2|$ 。

   Misunderstanding the properties of conjugate complex numbers: $z\cdot\bar{z}=|z|^2$ is a real number, but $|z_1z_2|=|z_1||z_2|$ and $|z_1+z_2|\le|z_1|+|z_2|$.

关键例题 Key Examples

例 1:当 m 为何值时,复数 $z=(2+i)m^2-3(1+i)m-2(1-i)$ 为(1)实数; (2)纯虚数?

Example 1: When does the complex number $z=(2+i)m^2-3(1+i)m-2(1-i)$ become (1) a real number; (2) a pure imaginary number?

解: $z=(2m^2-3m-2)+(m^2-3m+2)i$

Solution: $z=(2m^2-3m-2)+(m^2-3m+2)i$

(1) 实数: $m^2-3m+2=0\Rightarrow m=1$ 或 $m=2$;

(2) Real number: $m^2-3m+2=0\Rightarrow m=1$ or $m=2$;

(3) 纯虚数: $\begin{cases}2m^2-3m-2=0\\ m^2-3m+2\neq 0\end{cases}\Rightarrow m=-\frac{1}{2}$ 。

(4) Pure imaginary number: $\begin{cases}2m^2-3m-2=0\\ m^2-3m+2\neq 0\end{cases} \Longrightarrow m=-\frac{1}{2}$.

例 2(高考真题):设 $z=-3+2i$,则 $\overline{z}$ 在复平面内对应的点位于( )

Example 2 (College Entrance Examination Question): Let $z=-3+2i$, the point corresponding to $\overline{z}$ in the complex plane is located in ( )

A. 第一象限 B. 第二象限 C. 第三象限 D. 第四象限

A. First Quadrant B. Second Quadrant C. Third Quadrant D. Fourth Quadrant

解: $\overline{z}=-3-2i$,对应点 $(-3,-2)$,位于第三象限,选 C。

Solution: $\overline{z}=-3-2i$, corresponding to the point $(-3,-2)$, which is located in the third quadrant, choose C.

例 3:已知 x 是实数, y 是纯虚数,且 $(2x-1)+(3-y)i=y-i$,求 x 、 y 。

Example 3: Given that x is a real number and y is a pure imaginary number, and $(2x-1)+(3-y)i=y-i$, find x and y .

解:设 $y=bi$ ($b\in R$, $b\neq 0$),代入得 $(2x-1+b)+3i=(b-1)i$,故 $\begin{cases}2x-1+b=0\\ b-1=3\end{cases} \Longrightarrow x=-\frac{3}{2}$, $y=4i$ 。

Solution: Let $y=bi$ ($b\in R$, $b\neq 0$), substitute into the equation to get $(2x-1+b)+3i=(b >

7.2 复数的四则运算

Four Arithetic Operations of Complex Numbers

7.2.1 复数的加、减运算及其几何意义

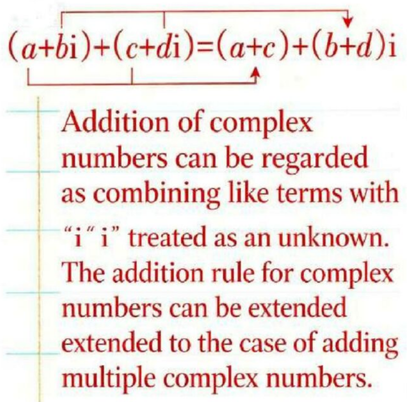
Addition and Subtraction of Complex Numbers and Their Geometric Meanings

核心知识归纳 Core Knowledge Summary

1. 运算法则 Operation Rules

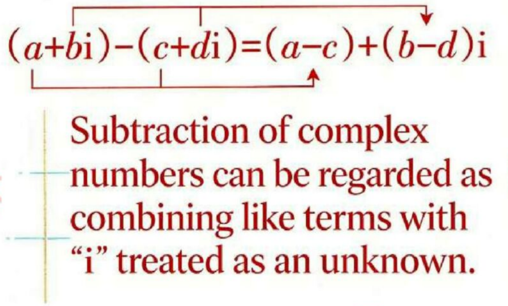
- 加法：$z_1+z_2=(a+c)+(b+d)i$ （$z_1=a+bi$，$z_2=c+di$),实部相加,虚部分别相加。

  Addition: $z_1+z_2=(a+c)+(b+d)i$ ($z_1=a+bi$, $z_2=c+di$), where real parts are added and imaginary parts are added respectively.



- 减法: $z_1-z_2=(a-c)+(b-d)i$,减法是加法的逆运算,实部相减,虚部分别相加。

  Subtraction: $z_1-z_2=(a-c)+(b-d)i$, subtraction is the inverse operation of addition, where real parts are subtracted and imaginary parts are subtracted respectively.



- 本质:将 i 看作"未知数",合并同类项。

  Essence: Treat i as an "unknown quantity" and combine like terms.

2. 运算律 Operational Laws

- 交换律: $z_1+z_2=z_2+z_1$ ；

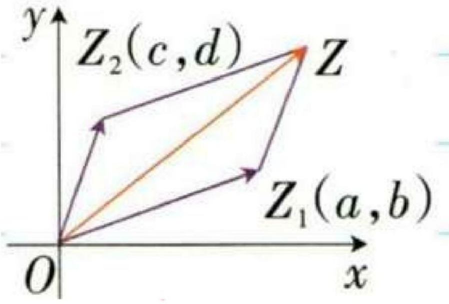
  Commutative Law: $z_1+z_2=z_2+z_1$;

- 结合律: $(z_1+z_2)+z_3=z_1+(z_2+z_3)$ 。

  Associative Law: $(z_1+z_2)+z_3=z_1+(z_2+z_3)$.

3. **几何意义 Geometric Meanings**

- 加法:对应向量的平行四边形法则(同起点)或三角形法则(首尾相接),与平面向量加法一致。

  Addition: Corresponding to the parallelogram rule (same starting point) or triangle rule (head-to-tail connection) of vectors, consistent with the addition of plane vectors.

- 减法:对应向量 $\overrightarrow{OZ_1}-\overrightarrow{OZ_2}=(a-c,b-d)$,即 $\overrightarrow{Z_2Z_1}$ (从 z_2 的对应点指向 z_1 的对应点)。

  Subtraction: Corresponding to the vector $\overrightarrow{OZ_1}-\overrightarrow{OZ_2}=(a-c,b-d)$, i.e., $\overrightarrow{Z_2Z_1}$ (from the corresponding point of z_2 to the corresponding point of z_1).

- 距离公式:复平面内两点 $Z_1(a,b)$ 、 $Z_2(c,d)$ 的距离 $|Z_1Z_2|=|z_1-z_2|=\sqrt{(a-c)^2+(b-d)^2}$ 。

  Distance Formula: The distance between two points $Z_1(a,b)$ and $Z_2(c,d)$ in the complex plane is $|Z_1Z_2|=|z_1-z_2|=\sqrt{(a-c)^2+(b-d)^2}$.

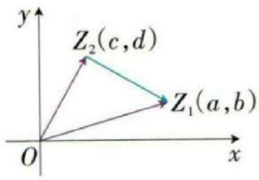
易错点解析 Common Mistakes Analysis

1. 虚部符号失误:减法运算中,虚部是 $b-d$,而非 $d-b$ (如 $(2+3i)-(1-2i)=1+5i$)。

   Mistake in the sign of the imaginary part: In subtraction, the imaginary part is $b-d$, not $d-b$ (e.g., $(2+3i)-(1-2i)=1+5i$).

2. 距离公式误区: $|z_1 - z_2|$ 表示两点间距离,而非 $|z_1| - |z_2|$ 。

   Misunderstanding the distance formula: $|z_1 - z_2|$ represents the distance between two points, not $|z_1| - |z_2|$.



3. 结果判断误区:两个虚数的差不一定是虚数(如 $(1+3i)-3i=1$)。

   Misjudgment of the result: The difference between two imaginary numbers is not necessarily an imaginary number (e.g., $(1+3i)-3i=1$).

关键例题 Key Examples

例 1(高考真题):设复数 z 满足 $|z-i|=1$, z 在复平面内对应的点为 (x,y) ,则( )

Example 1 (College Entrance Examination Question): Let the complex number z satisfy $|z-i|=1$, and the point corresponding to z in the complex plane is (x,y) , then ( )

A. $(x+1)^2+y^2=1$ B. $(x-1)^2+y^2=1$ C. $x^2+(y-1)^2=1$ D. $x^2+(y+1)^2=1$

解: $z=x+yi$, $|z-i|=|x+(y-1)i|=\sqrt{x^2+(y-1)^2}$,故 $x^2+(y-1)^2=1$,选 C。

Solution: $z=x+yi$, $|z-i|=|x+(y-1)i|=\sqrt{x^2+(y-1)^2}$, so $x^2+(y-1)^2=1$, choose C.

例 2:计算 $(5-6i)+(-2-i)-(3+4i)=-11i$ 。

Example 2: Calculate $(5-6i)+(-2-i)-(3+4i)=-11i$.

7.2.2 复数的乘、除运算 Multiplication and Division of Complex Numbers

核心知识归纳 Core Knowledge Summary

1. **乘法法则: $\boldsymbol{z_1z_2=(a+bi)(c+di)=(ac-bd)+(ad+bc)i}$ 。**

   Multiplication Rule: $z_1z_2=(a+bi)(c+di)=(ac-bd)+(ad+bc)i$.

   - 本质:多项式相乘, $i^2=-1$,合并实部和虚部。

     Essence: Multiply like polynomials, substitute $i^2=-1$, and combine real and imaginary parts.

- 运算律:交换律 $z_1z_2 = z_2z_1$ ；结合律 $(z_1z_2)z_3 = z_1(z_2z_3)$ ；分配律 $z_1(z_2+z_3) = z_1z_2 + z_1z_3$ 。

  Operational Laws: Commutative Law $z_1z_2 = z_2z_1$; Associative Law $(z_1z_2)z_3 = z_1(z_2z_3)$; Distributive Law $z_1(z_2+z_3) = z_1z_2 + z_1z_3$.

- 常用公式(平方差、完全平方):

  Common Formulas (Difference of Squares, Perfect Squares):

  - $(a+bi)(a-bi) = a^2+b^2$ (共轭复数相乘为实部平方加虚部平方);

    $(a+bi)(a-bi) = a^2+b^2$ (the product of conjugate complex numbers is the sum of the square of the real part and the square of the imaginary part);

  - $(a \pm bi)^2 = a^2 - b^2 \pm 2abi$ 。

2. **除法法则:除法是乘法的逆运算，$z_1 \div z_2 = \frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$ ($c+di \neq 0$)。**

Division Rule: Division is the inverse operation of multiplication, $z_1 \div z_2 = \frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$ ($c+di \neq 0$).

- 关键:分子分母同乘分母的共轭复数 $c-di$,实现分母“实数化”(类似根式除法的分母有理化)。

  Key: Multiply both the numerator and denominator by the conjugate complex number $c-di$ of the denominator to “rationalize the denominator” (similar to rationalizing the denominator in radical division).

3. **常用结论 Common Conclusions**

- i 的幂的周期性: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^{4n+k} = i^k$ ($n \in N^*$, $k = 1,2,3$); $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$ 。

Periodicity of Powers of i : $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^{4n+k} = i^k$ ($n \in N^*$, $k = 1,2,3$); $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$.

- 特殊复数 $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ 的性质: $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $\overline{\omega} = \omega^2$ 。

  Properties of the Special Complex Number $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$: $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $\overline{\omega} = \omega^2$.

4. **复数范围内的一元二次方程 Univariate Quadratic Equations in the Complex Number Field**

- 实系数方程 $ax^2 + bx + c = 0$ ($a \neq 0$):

  Real-Coefficient Equation $ax^2 + bx + c = 0$ ($a \neq 0$):

  - $\Delta = b^2 - 4ac \geq 0$:实根 $x = \frac{-b \pm \sqrt{\Delta}}{2a}$;
  - $\Delta = b^2 - 4ac \geq 0$: Real roots $x = \frac{-b \pm \sqrt{\Delta}}{2a}$;
  - $\Delta < 0$:虚根 $x = \frac{-b \pm \sqrt{-\Delta}}{2a}i$ (共轭虚根)。
  - $\Delta < 0$: Imaginary roots $x = \frac{-b \pm \sqrt{-\Delta}}{2a}i$ (conjugate imaginary roots).

- 性质:实系数方程的虚根成对出现; 复系数方程无此性质。

  Property: The imaginary roots of real-coefficient equations appear in pairs; this property does not hold for complex-coefficient equations.

易错点解析 Common Mistakes Analysis

1. 乘法运算中 i^2 失误:忘记将 i^2 化为 -1 (如 $(1+i)(2+i) = 1 + 3i$,而非 $2 + 3i$)。

   Mistake in i^2 during multiplication: Forgetting to substitute $i^2 = -1$ (e.g., $(1+i)(2+i) = 1 + 3i$, not $2 + 3i$).

2. 除法分母实数化误区:分母乘自身共轭复数,而非分子的共轭复数。

Misunderstanding of rationalizing the denominator in division: Multiply the denominator by its own conjugate complex number, not the conjugate complex number of the numerator.

3. 幂的周期性误用:忽略 i 的幂周期为 4 ,直接硬算高次幂。

   Misuse of the periodicity of powers: Ignoring that the period of powers of i is 4 and directly calculating high powers.

4. 复系数方程误区:将实系数方程的虚根成对性质推广到复系数方程。

   Misunderstanding of complex-coefficient equations: Extending the property of conjugate imaginary roots of real-coefficient equations to complex-coefficient equations.

关键例题 Key Examples

例 1(高考真题):已知 $z=2+i$,则 $z\cdot\bar{z}=$ ( )

Example 1 (College Entrance Examination Question): Given $z=2+i$, then $z\cdot\bar{z}=$ ( )

A. $\sqrt{3}$ B. $\sqrt{5}$ C. 3 D. 5

解: $z\cdot\bar{z}=|z|^2=2^2+1^2=5$,选 D。

Solution: $z\cdot\bar{z}=|z|^2=2^2+1^2=5$, choose D.

例 2(高考真题):若 $z(1+i)=2i$,则 $z=$ ( )

Example 2 (College Entrance Examination Question): If $z(1+i)=2i$, then $z=$ ( )

A. $-1-i$ B. $-1+i$ C. $1-i$ D. $1+i$

解: $z=\frac{2i}{1+i}=\frac{2i(1-i)}{(1+i)(1-i)}=\frac{2+2i}{2}=1+i$,选 D。

Solution: $z=\frac{2i}{1+i}=\frac{2i(1-i)}{(1+i)(1-i)}=\frac{2+2i}{2}=1+i$, choose D.

例 3:解方程 $x^2+x+1=0$ (复数范围内)。

Example 3: Solve the equation $x^2+x+1=0$ (in the complex number field).

解: $\varDelta = 1 - 4 = -3 < 0$,根为 $x = \frac{-1 \pm \sqrt{3}i}{2}$ 。

Solution: $\varDelta = 1 - 4 = -3 < 0$, the roots are $x = \frac{-1 \pm \sqrt{3}i}{2}$.